

# Ricardo Manuel da Encarnação dos Santos

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## Personal Summary

**Senior Software Engineer** with deep expertise in C#, ASP.NET Core, and Python, specializing in cross-platform backend services, distributed systems, and real-time telephony APIs.

**Production AI/ML Experience:** Track record of architecting scalable, multi-tenant Kubernetes services to train, deploy, and integrate NLP/deep learning models into live enterprise systems.

**Cloud-Native Operations & Ownership:** End-to-end expertise in building resilient cloud-native ecosystems on Kubernetes, featuring robust observability (OpenTelemetry, Prometheus, Grafana), automated CI/CD pipelines, and multi-cloud/on-prem deployments.

## Professional Experience

### Senior Software Engineer

Enghouse Interactive



**Feb 2022 - Present | Lisbon, Portugal (Remote, Full-time)**

- **Served as lead developer** for a cross-platform UCaaS (Unified Communications as a Service) backend service, guiding the technical implementation of a unified API that bridges multiple telephony platforms across Linux, Windows, Kubernetes, and on-premises environments.
- **Engineered robust server-side applications** using C#, ASP.NET Core, Entity Framework Core, PostgreSQL, and Python within a geographically distributed engineering team.
- **Extended telephony API with generative AI** (Gemini) capabilities, enabling the creation and management of autonomous call center agents with native tool and function-calling support for agentic workflows.
- **Integrated enterprise observability** by embedding OpenTelemetry collectors to export traces, logs, and metrics into a Prometheus and Grafana stack, reducing system troubleshooting time.
- **Implemented CI pipelines** using Jenkins and Bitbucket to automate building, testing, and exporting new releases. Implemented automated tests, including unit tests and functional tests using xUnit and Microsoft Test Framework.
- **Experience with AI-assisted development** by daily usage of GitHub Copilot integrated in Visual Studio and Visual Studio Code.

### Software Engineer

Enghouse Interactive



**Jan 2021 - Jan 2022 | Lisbon, Portugal (Remote, Full-time)**

- **Developed a web-based, low-code IVR flow designer** using ASP.NET/C# and React/Redux/Typescript, enabling non-technical users to build complex inbound logic.

## Core Skills & Technologies

### Languages

- C#
- Python
- SQL
- Java
- TypeScript

### Frameworks

- .NET / ASP.NET Core
- Entity Framework Core
- Windows Forms

### Cloud & DevOps

- Docker
- Kubernetes
- Git / GitLab / Bitbucket
- Jenkins
- Azure ML
- Google Cloud
- Apache Kafka

### Databases

- PostgreSQL
- MongoDB
- SQL Server

### Observability

- OpenTelemetry
- Prometheus
- Grafana

### Architecture

- REST APIs
- Microservices
- Event-driven architecture

### Engineering Practices

- Agile / Scrum
- AI-assisted development (*GitHub Copilot*)

## Languages

- Portuguese: Native

- **Integrated with a Natural Language Processing (NLP) engine**, incorporating third-party services (Microsoft LUIS) to automatically detect the intent of customer calls and route the call to the appropriate flow.

- **English:** Professional Working Proficiency (B2)

## AI & Backend Engineer

Altitude Software

May 2015 - Jan 2021 | Lisbon, Portugal (Hybrid, Full-time)



- **Designed and implemented scalable, multi-tenant Kubernetes microservices** using Python and Azure Machine Learning HyperDrive to automate the end-to-end training and deployment of neural network models.
- **Built an autonomous live-chat recommendation engine** that leveraged machine learning models trained on conversation transcripts to provide predictive, context-aware recommendations to live agents in real time. Automated knowledge retrieval for live-chat agents, accelerating response times and promoting consistent answers across the contact center.
- **Researched predictive dialing using deep reinforcement learning**, developing and evaluating prototypes based on DDPG, RDPG, and Recurrent Q-Learning in Python, and extending the Java-based JaamSim simulation platform to validate their performance.
- **Owned production-grade microservices end to end** within a distributed system, with hands-on experience across the Azure ecosystem, MongoDB Atlas, Apache Kafka, and observability tooling including Graylog, Prometheus, Jaeger, and Grafana.

## Software Developer

Altitude Software

Jun 1997 - Apr 2015 | Lisbon, Portugal (On-site, Full-time)



- **Designed and developed uAgent Windows**, Altitude's flag-ship Windows desktop client, supporting high-concurrency voice, email, and live-chat agent interactions.
- **Architected .NET APIs and a layout editor** enabling Professional Services to build third-party integrations, embedded controls, and tailored workflow automation.
- **Implemented a high-performance call routing engine** in collaboration with Alcatel-Lucent, delivering complex, real-time call distribution logic for enterprise telephony systems.

## Junior Web Developer

Megamedia

Oct 1996 - May 1997 | Lisbon, Portugal (On-site, Full-time)



Developed web design elements and client-facing interfaces for an early enterprise online banking platform (Santander-Totta).

## Education

Post graduation, Advanced Management Program for Executives  
Católica Lisbon School of Business and Economics | 2001 - 2002



Licentiate Degree in Information Systems and Computer Engineering  
Instituto Superior Técnico (University of Lisbon) | 1991 - 1996



- **Specialization:** Information Systems and Databases | **Grade: 15.3**
- **Final Project:** Designed and built a 3D Virtual Desktop application managing Windows files and applications in an interactive 3D environment.
- **Leadership:** Member of the Board of Directors, IST Student's Union (1995 - 1996).

## Additional Training

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Machine Learning    Coursera / Stanford Online (Andrew Ng) | 2018

**Stanford** | ONLINE

Keras: Deep Learning in Python    Udemy | 2017

Machine Learning A-Z: Hands-On Python & R In Data Science    Udemy | 2017

Complete Python Bootcamp    Udemy | 2017

Usability in Practice (Methodologies)    Nielsen Norman Group | 2007